

Mohamed Mamdouh

Full-Stack Developer | .NET | Angular

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SUMMARY

Full-Stack .NET & Angular Developer with a solid foundation in building end-to-end web apps. Specialized in robust backend systems using ASP.NET Core and Clean Architecture, paired with modular Angular frontends. A proactive problem-solver ready to deliver clean, scalable, and testable code.

EDUCATION

Bachelor of Engineering – Computer and Systems Engineering (Zagazig University).

TECHNICAL SKILLS

Backend: C#, ASP.NET Core Web API, Entity Framework Core, REST APIs, ASP.NET Identity, JWT Authentication, Role-Based Authorization, ASP.NET Identity, Middleware, Clean Architecture, DDD, CQRS, MediatR, SignalR, FluentValidation, Background Services, Caching, Dependency Injection, Unit & Integration Testing.

Frontend: Angular, TypeScript, JavaScript, HTML, CSS, Bootstrap, Tailwind CSS.

Database: SQL, T-SQL, ADO.NET, MongoDB, Database Design, Query Optimization.

Tools: Git, GitHub, Docker, Postman, Swagger, CI/CD basics.

Concepts: OOP, SOLID Principles, Design Patterns, Clean Code, Problem Solving, SDLC.

SELECTED PROJECTS

Amanah – Lost & Found Platform

Full-stack production lost-and-found platform, handling report submissions, moderation queues, ownership claims, and real-time chat.

[GitHub](#) [Live Production](#)

Tech Stack: .NET, Angular (RTL/Arabic), PostgreSQL, EF Core, FluentValidation, JWT Bearer, SignalR, Cloudflare R2, Docker, xUnit, Testcontainers.

Key Highlights

- **Monolith Architecture:** Built a full-stack monorepo with an ASP.NET Core API serving an Angular SPA from wwwroot for single-origin deployment.
- **Secure Auth & Phone OTP:** Implemented phone OTP signup, password sign-in, and short-lived JWTs paired with httpOnly refresh cookies.
- **Reports & Ownership Claims:** Built report submission flows with EXIF stripping/WebP conversion, admin moderation queues, and private photo claims.
- **Real-Time Messaging & Outbox:** Integrated SignalR for live chat, backed by REST fallbacks and transactional outboxes for reliable SMS and email dispatch.
- **Background Jobs & Resilience:** Included background lifecycle workers for listing expirations, claim timeouts, orphaned storage sweeps, and rate limiting policies.
- **Automated Testing & CI/CD:** Wrote integration tests using xUnit and Testcontainers with automated CI/CD and keepalive workflows via GitHub Actions.

Shora – Full-Stack Consulting & Booking Platform

[GitHub](#) [Live Production](#)

Full-stack production booking platform for one-to-one consulting sessions with availability management, manual payment verification, client bookings, and admin operations.

Tech Stack: .NET 10, Angular, PostgreSQL, EF Core, ASP.NET Core Identity, JWT Bearer, Cloudflare R2, Docker, xUnit, Testcontainers, GitHub Actions.

Key Highlights

- **Clean Architecture:** Built a full-stack app using .NET and Angular with a clear separation of business logic, UI.
- **Secure Authentication:** Added JWT login, httpOnly refresh token rotation, email verification, and role-based authorization.
- **Booking & Payment Pipeline:** Created booking flows with slot holds, manual payment receipt uploads, admin approval, and cancellation management.
- **Private Storage & Outbox Email:** Stored receipt uploads securely in Cloudflare R2 with presigned URLs and built a transactional email outbox system using Brevo.
- **Background Jobs & Resilience:** Added background workers to release expired slot holds, auto-complete past bookings, rate limit APIs, and run health monitors.
- **Automated Testing & CI/CD:** Wrote integration tests using xUnit and Testcontainers, and automated deploys via GitHub Actions and Docker on Render.

MechanicShop – Workshop Operations API

Backend REST API for a mechanic shop covering work orders, day scheduling, labor assignment, customer and vehicle records, and invoicing.

Tech Stack: ASP.NET Core, SQL Server, EF Core, MediatR, JWT, ASP.NET Identity, FluentValidation, Docker, xUnit, Testcontainers, GitHub Actions.

Key Highlights

- Built a REST API using ASP.NET Core and Clean Architecture with CQRS and MediatR.
- Implemented JWT authentication, refresh-token rotation, device sessions, and role-based authorization.
- Built work-order and scheduling workflows with labor assignment, shop-day management, repair tasks, and invoicing.
- Added unit and integration tests using xUnit and Testcontainers, with automated CI/CD through GitHub Actions.

DVLD – Driver & Vehicle License Department System

Desktop application for managing driver licenses, license applications, driving tests, and vehicle-related records.

Tech Stack: C#, .NET Framework, Windows Forms, ADO.NET, SQL Server.

Key Highlights

- Built a desktop application using 3-tier architecture with clear separation of concerns.
- Implemented driver, person, user, and license management with role-based access control.
- Developed workflows for license issuance, renewal, replacement, international licenses, and license detention.
- Added driving test scheduling, appointment management, and exam question bank features.